

	Fecha	Tasa_participacion
4	2005-01-01	40.351692
5	2005-02-01	40.432370
6	2005-03-01	41.179709
7	2005-04-01	39.732705
8	2005-05-01	41.027773
...	...	...
253	NaT	NaN
254	NaT	NaN
255	NaT	NaN
256	NaT	NaN
257	NaT	NaN

254 rows × 2 columns

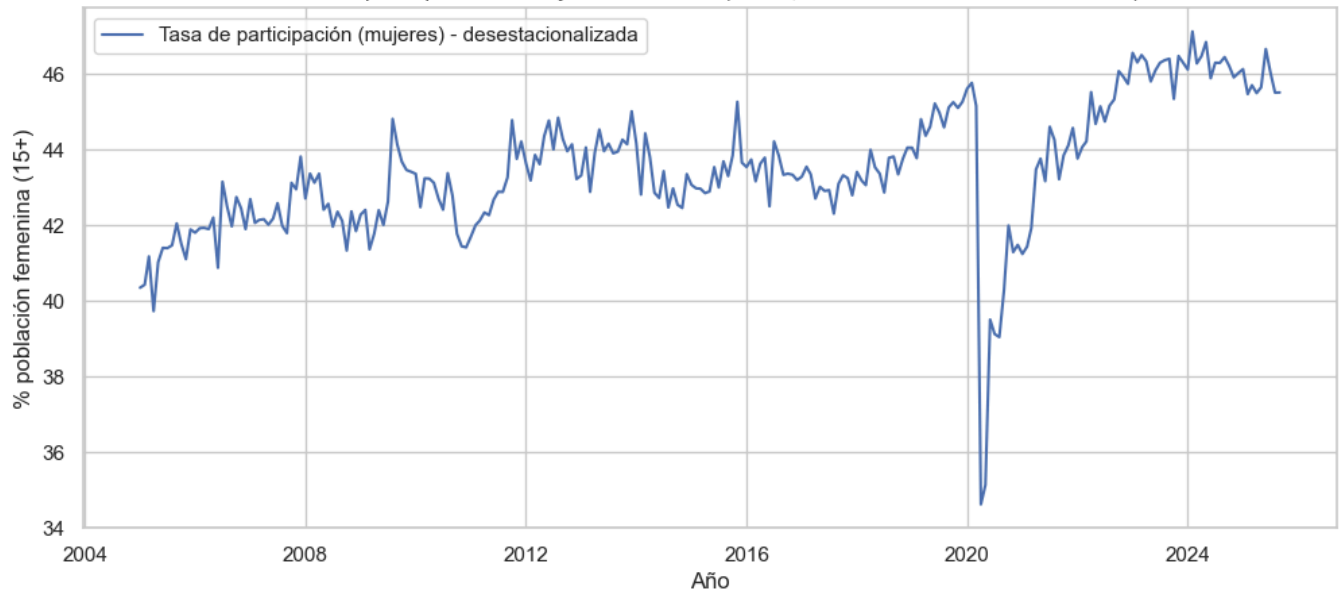
Periodo detectado: 2005-01 a 2025-09

Resumen estadístico:

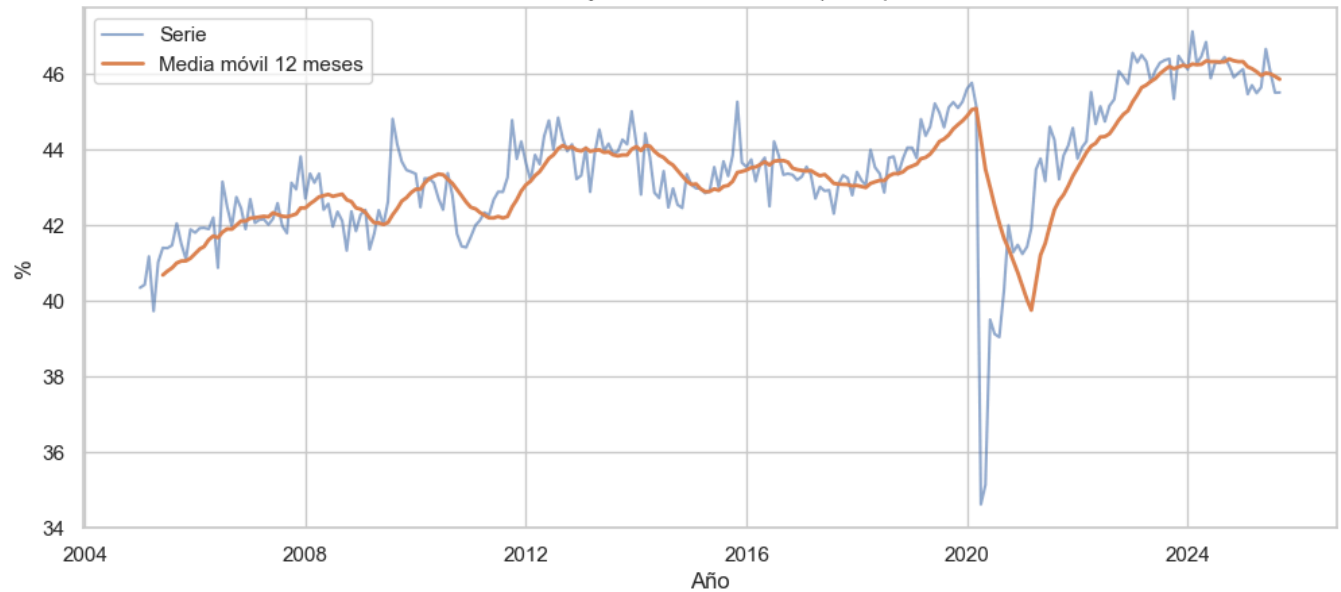
```
count    249.000000
mean      43.472531
std        1.748154
min       34.616667
25%       42.457499
50%       43.366709
75%       44.539689
max       47.135221
```

Archivo limpio guardado como: tasa_participacion_mujeres_clean.csv

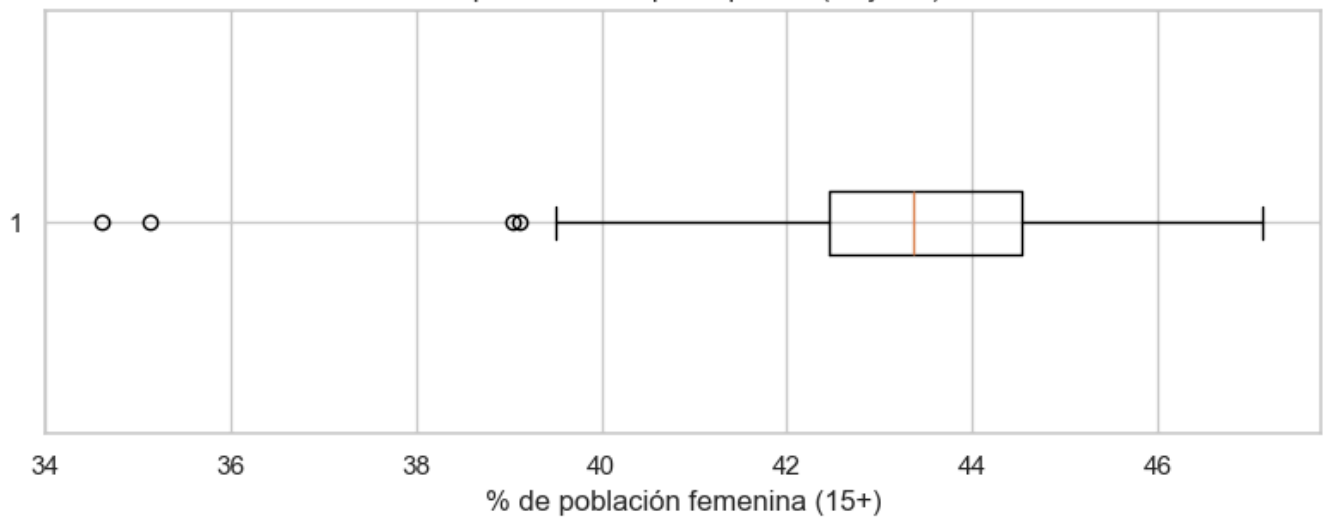
Tasa de participación de mujeres en México (INEGI, ENOE - serie desestacionalizada)



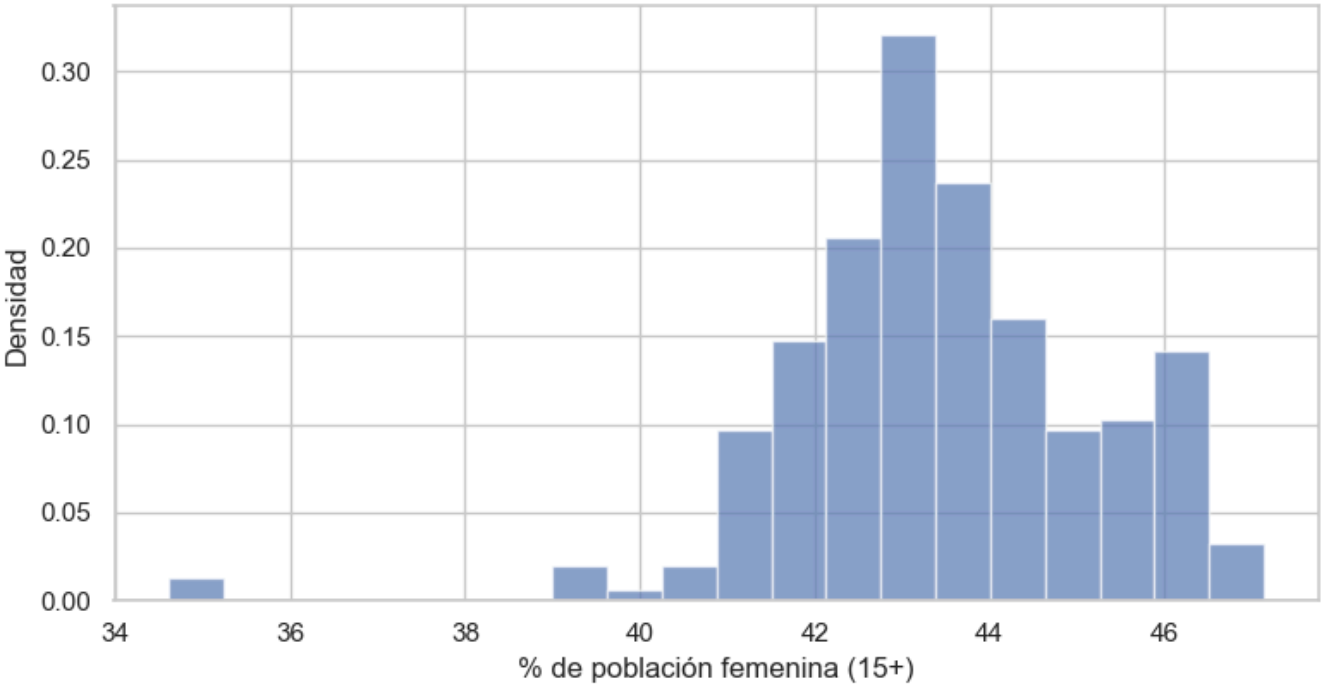
Serie y tendencia suavizada (MA12)



Boxplot - Tasa de participación (mujeres)



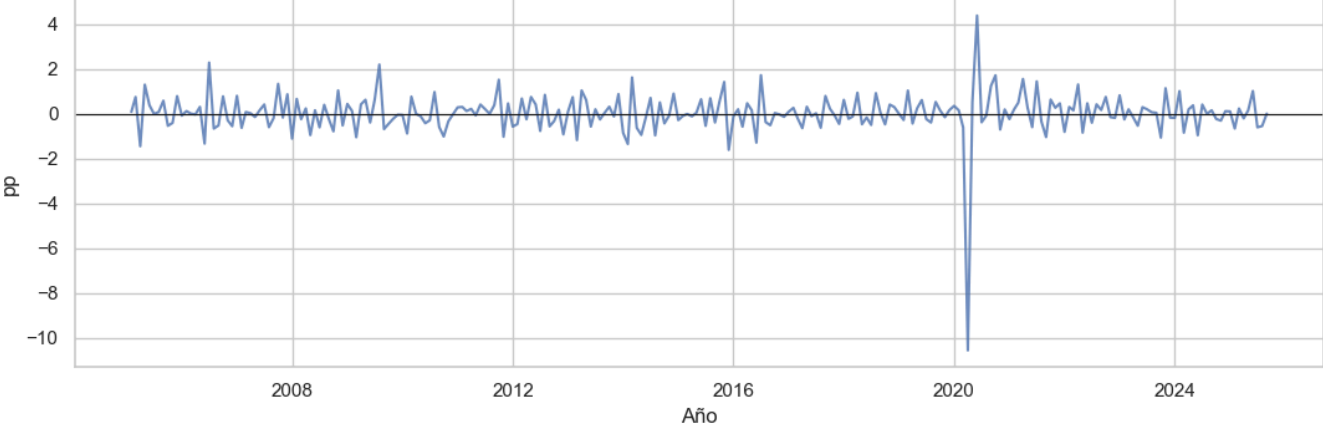
Distribución de la tasa de participación femenina

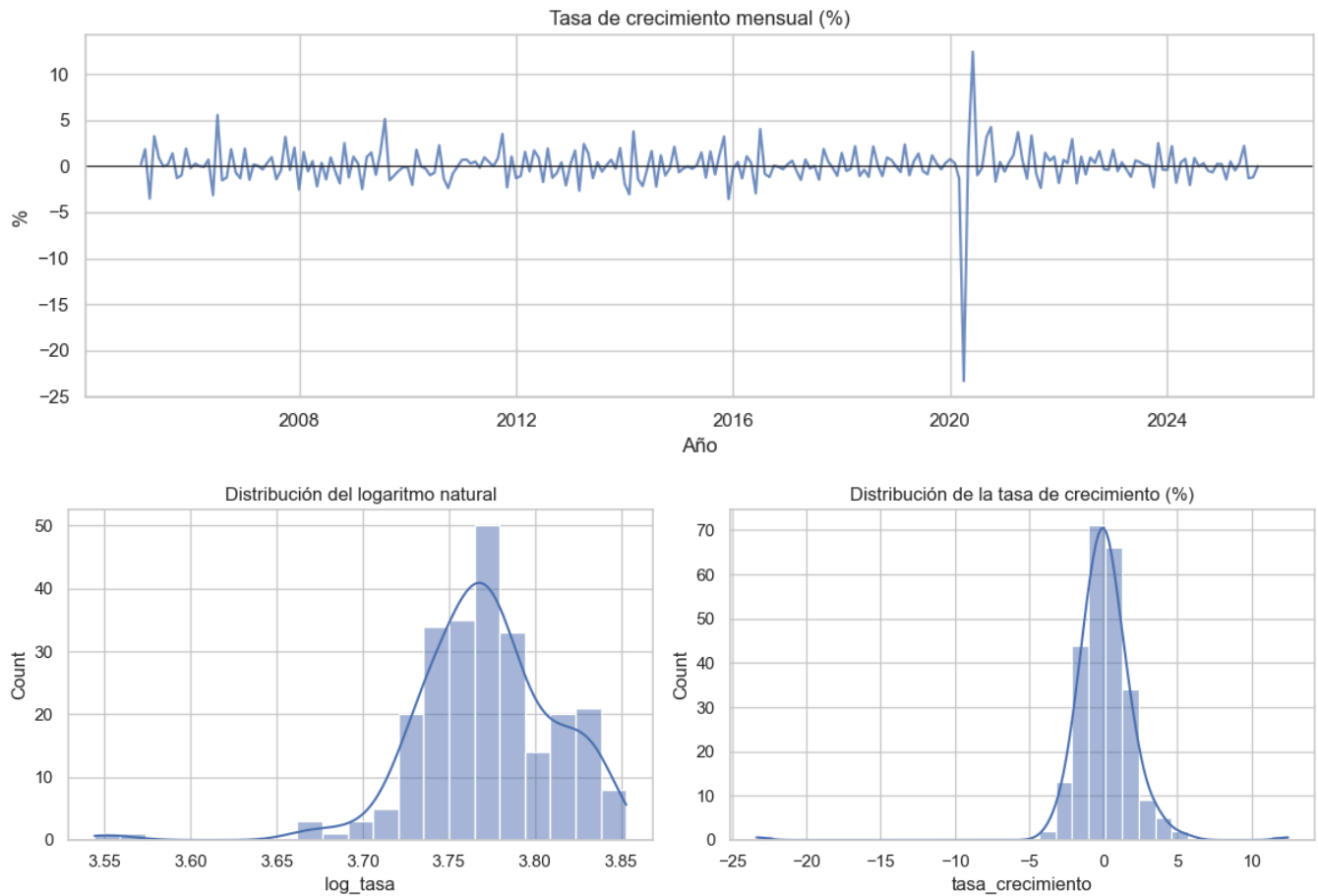


Variabilidad temporal: desviación estándar en el tiempo (12m)



Diferencia mensual (puntos porcentuales)





(Sección 2) Estadísticos y exploración

Periodo analizado: 2005-01 – 2025-09 | Observaciones: 249

Media: 43.47 %, Desv.Est.: 1.75 pp, Min: 34.62 %, Mediana: 43.37 %, Max: 47.14 %

Asimetría (skew): -0.819 | Curtosis (Fisher): 3.619

Posibles outliers (IQR): 4 observación(es) (ver meses listados abajo).

Meses marcado(s) como posible(s) outlier por IQR:

Fecha

2020-04-01 34.616667

2020-05-01 35.132391

2020-07-01 39.125171

2020-08-01 39.044937

Prueba Dickey-Fuller Aumentada (ADF) - Serie en niveles

ADF Statistic: -3.132231476947229

p-value: 0.024258055850297087

Lags Used: 2

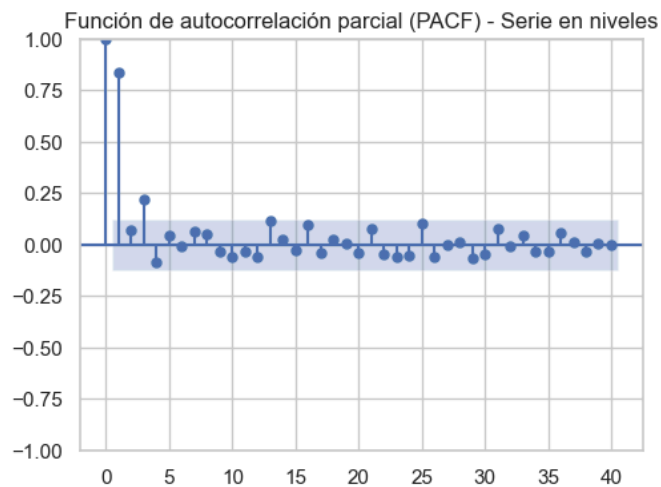
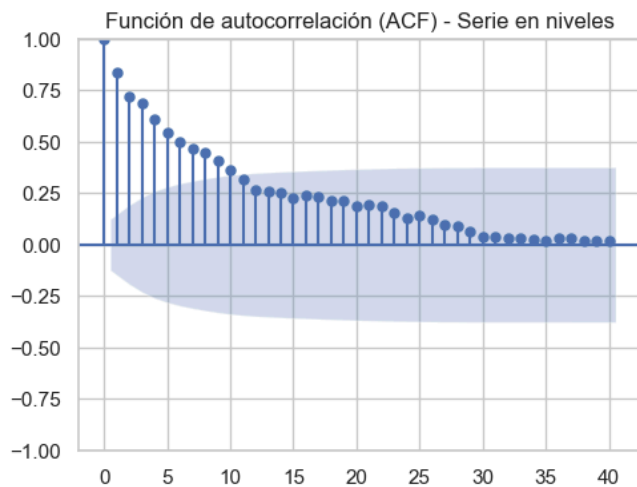
Observations: 246

Critical Value 1%: -3.457215237265747

Critical Value 5%: -2.873361841566324

Critical Value 10%: -2.5730700760129555

Conclusión: Rechazar H_0 (serie estacionaria)



Prueba Dickey-Fuller Aumentada (ADF) - Primera diferencia

ADF Statistic: -15.902705355748825

p-value: 8.293961936443131e-29

Lags Used: 1

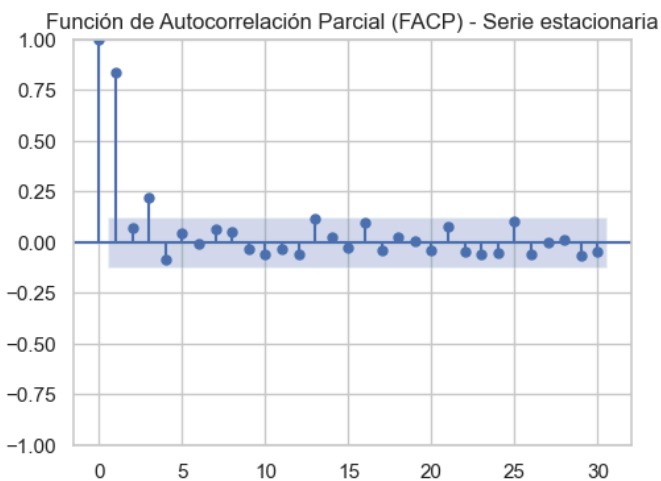
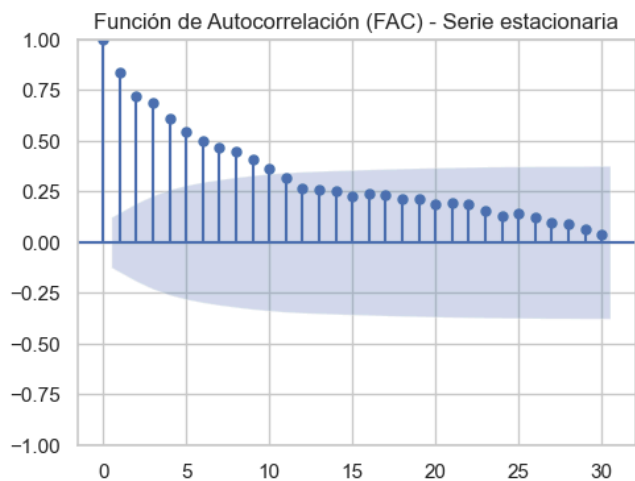
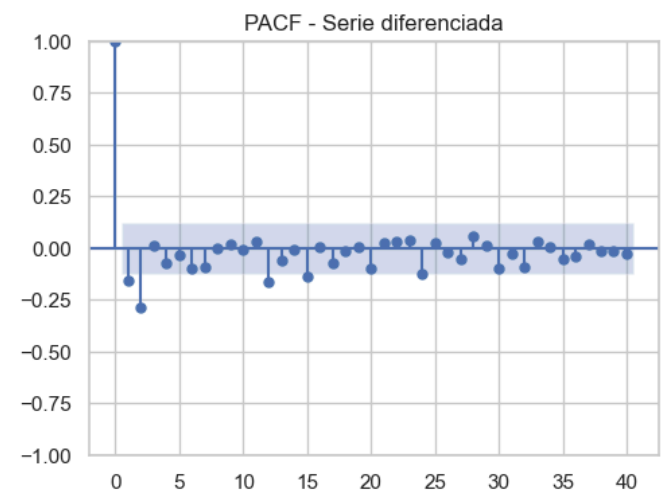
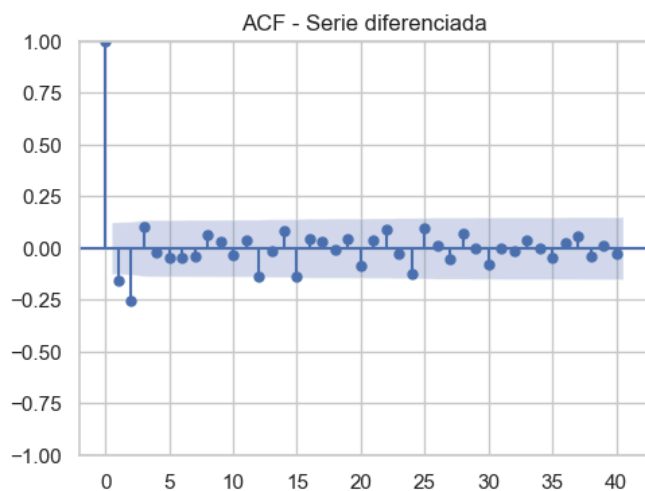
Observations: 246

Critical Value 1%: -3.457215237265747

Critical Value 5%: -2.873361841566324

Critical Value 10%: -2.5730700760129555

Conclusión: Rechazar H_0 (serie estacionaria)



Primeros coeficientes de la FAC:

Lag 0: 1.000
Lag 1: 0.836
Lag 2: 0.720
Lag 3: 0.684
Lag 4: 0.608
Lag 5: 0.546
Lag 6: 0.500
Lag 7: 0.468
Lag 8: 0.449
Lag 9: 0.411
Lag 10: 0.360

Primeros coeficientes de la FACP:

Lag 0: 1.000
Lag 1: 0.839
Lag 2: 0.072
Lag 3: 0.227
Lag 4: -0.086
Lag 5: 0.045
Lag 6: -0.011
Lag 7: 0.070
Lag 8: 0.052
Lag 9: -0.033
Lag 10: -0.062

=== Modelo AR(1) estimado sobre la serie en niveles ===

SARIMAX Results

```
=====
Dep. Variable:      Tasa_participacion    No. Observations:      249
Model:              ARIMA(1, 0, 0)        Log Likelihood        -337.648
Date:              Sat, 15 Nov 2025      AIC                   681.296
Time:              17:51:49              BIC                   691.848
Sample:            01-01-2005            HQIC                  685.543
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
const         43.4494      0.506      85.895      0.000      42.458      44.441
ar.L1          0.8480      0.031      27.307      0.000       0.787       0.909
sigma2         0.8773      0.024      36.911      0.000       0.831       0.924
=====
Ljung-Box (L1) (Q):              1.42    Jarque-Bera (JB):              34836.33
Prob(Q):                        0.23    Prob(JB):                  0.00
Heteroskedasticity (H):          3.80    Skew:                      -5.10
Prob(H) (two-sided):            0.00    Kurtosis:                  60.04
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Parámetros principales:

Constante: 43.4494

Phi_1 (AR1): 0.8480

AIC: 681.30

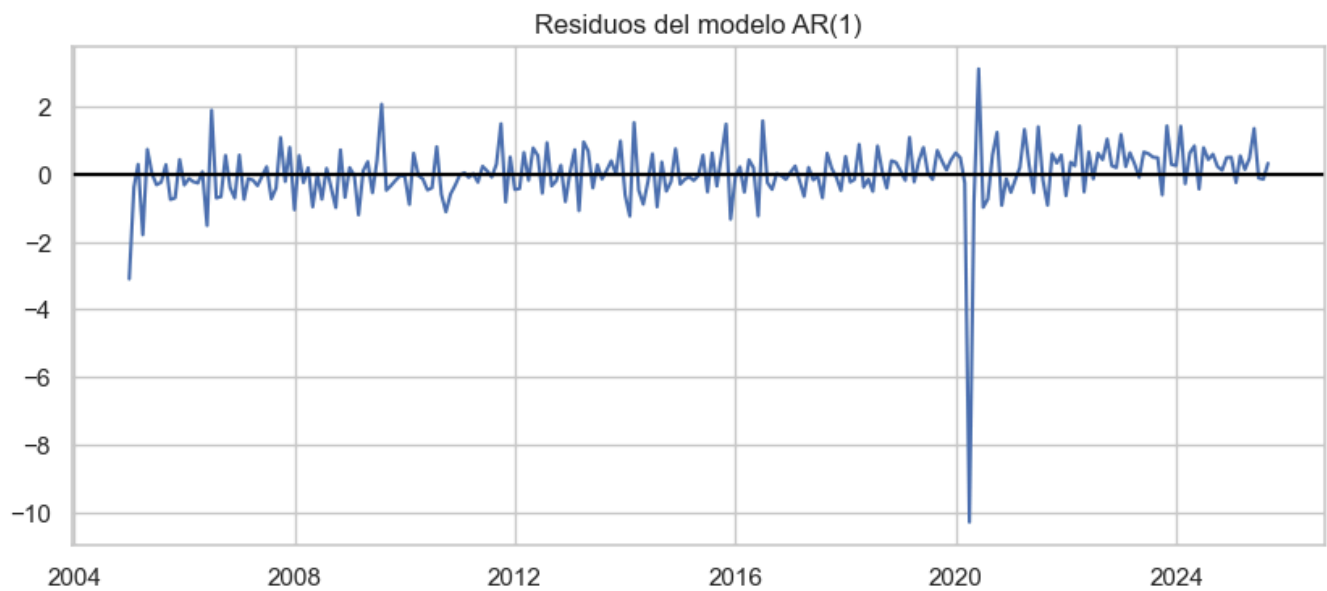
BIC: 691.85

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
```

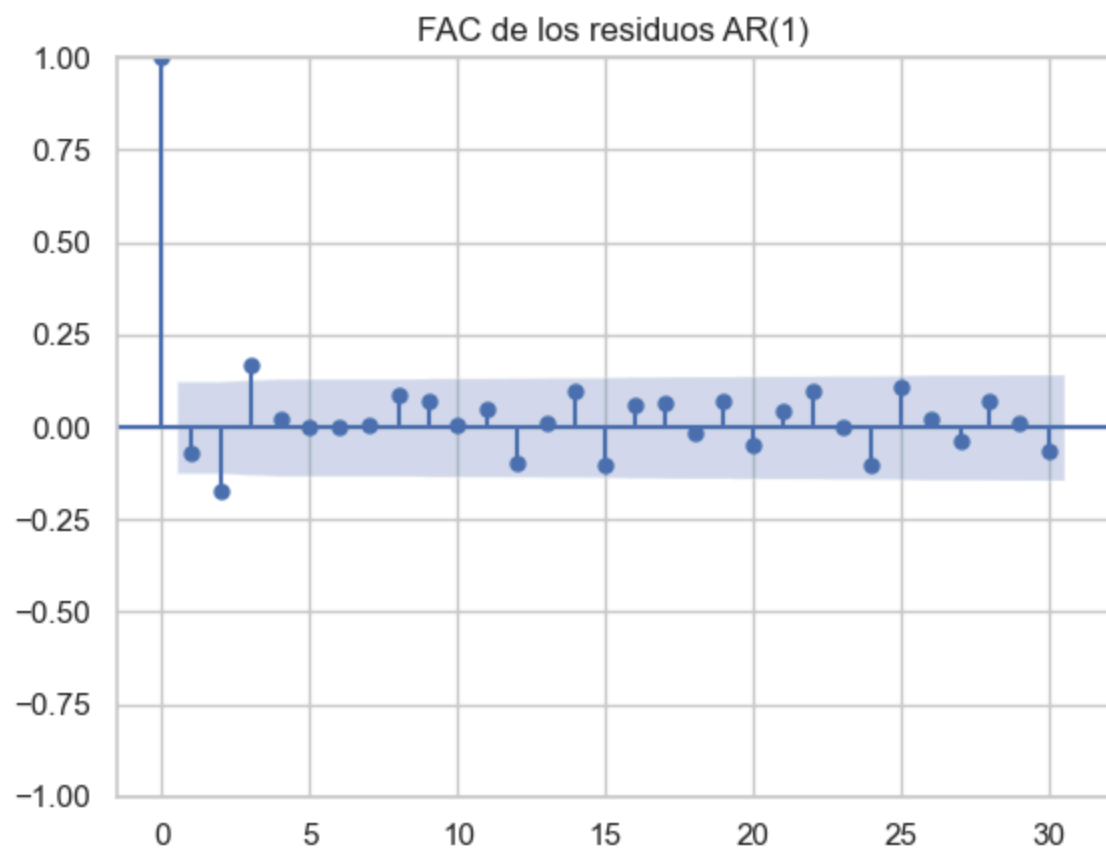
=== Prueba de Ljung-Box (residuos AR(1)) ===

```
      lb_stat  lb_pvalue
10  19.270314   0.036961
```

Conclusión: Se rechaza H_0 ; los residuos muestran autocorrelación (modelo AR(1) puede ser insuficiente).

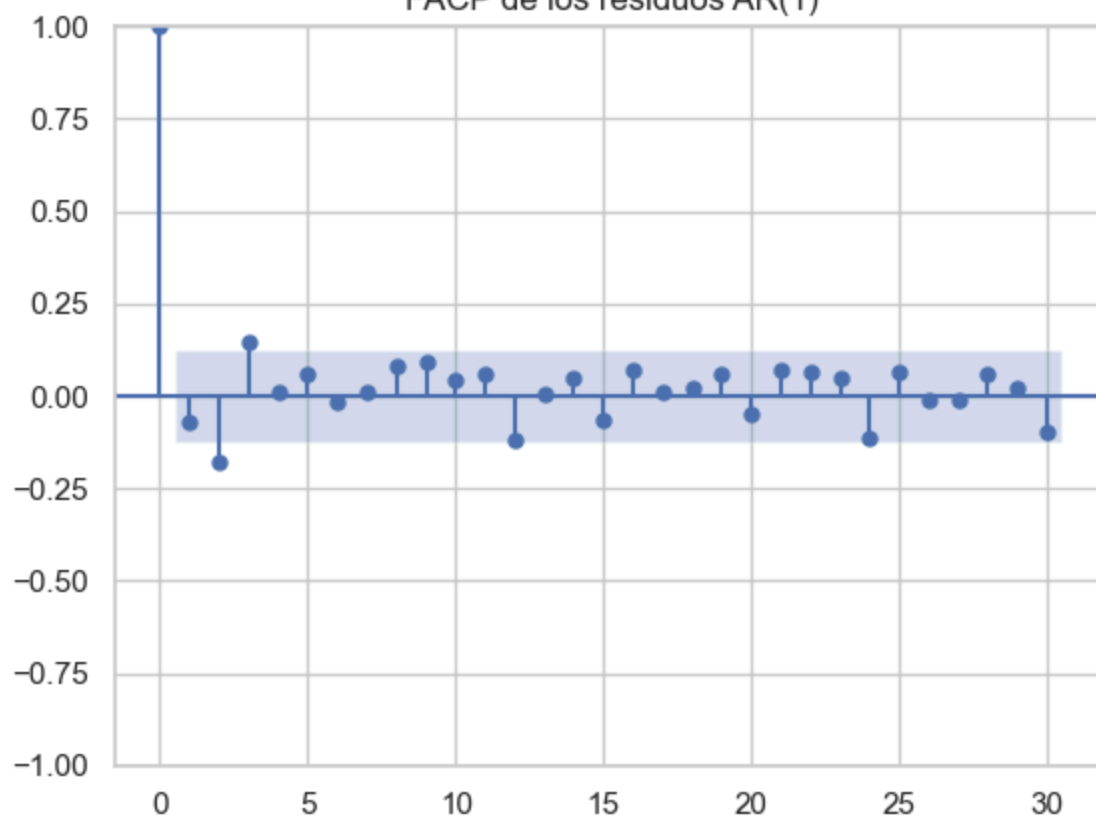


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<Figure size 800x400 with 0 Axes>

FACP de los residuos AR(1)



=== Modelo MA(1) estimado sobre la serie en niveles ===

SARIMAX Results

```
=====
Dep. Variable:      Tasa_participacion    No. Observations:      249
Model:              ARIMA(0, 0, 1)        Log Likelihood        -399.829
Date:               Sat, 15 Nov 2025      AIC                    805.658
Time:               18:13:53              BIC                    816.211
Sample:             01-01-2005            HQIC                   809.906
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
const         43.4728      0.151     288.085      0.000      43.177      43.769
ma.L1          0.7455      0.039     19.339      0.000       0.670       0.821
sigma2         1.4482      0.053     27.331      0.000       1.344       1.552
=====
```

```
=====
Ljung-Box (L1) (Q):      33.40    Jarque-Bera (JB):      2056.62
Prob(Q):                 0.00    Prob(JB):              0.00
Heteroskedasticity (H):   2.81    Skew:                -1.87
Prob(H) (two-sided):     0.00    Kurtosis:             16.57
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Parámetros principales:

Constante: 43.4728

Theta_1 (MA1): 0.7455

AIC: 805.66

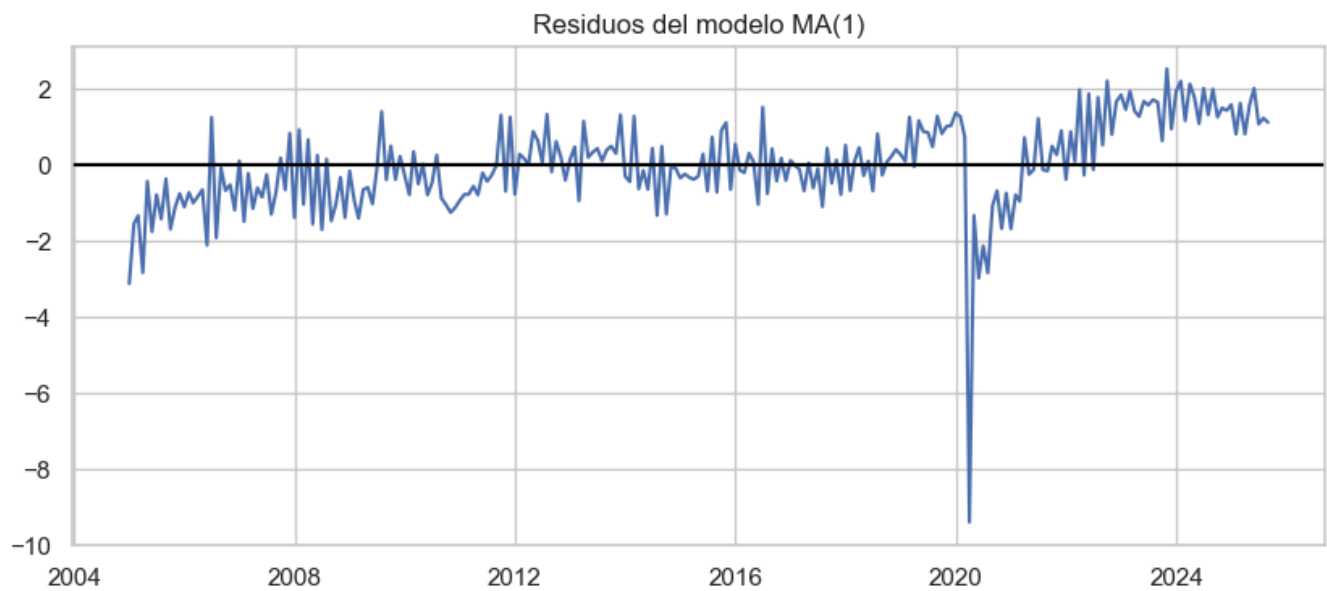
BIC: 816.21

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
```

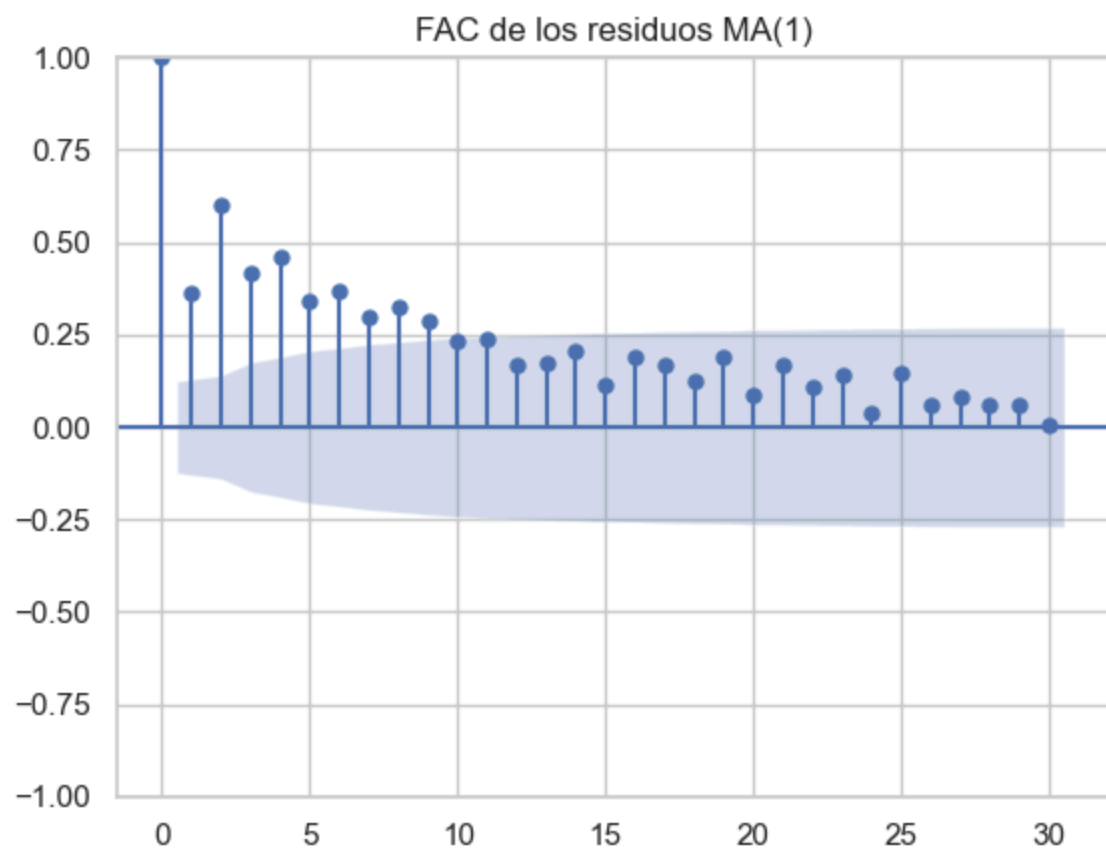
=== Prueba de Ljung-Box (residuos MA(1)) ===

```
      lb_stat      lb_pvalue
10  373.149588  4.832342e-74
```

Conclusión: Se rechaza H_0 ; los residuos muestran autocorrelación (modelo MA(1) puede ser insuficiente).

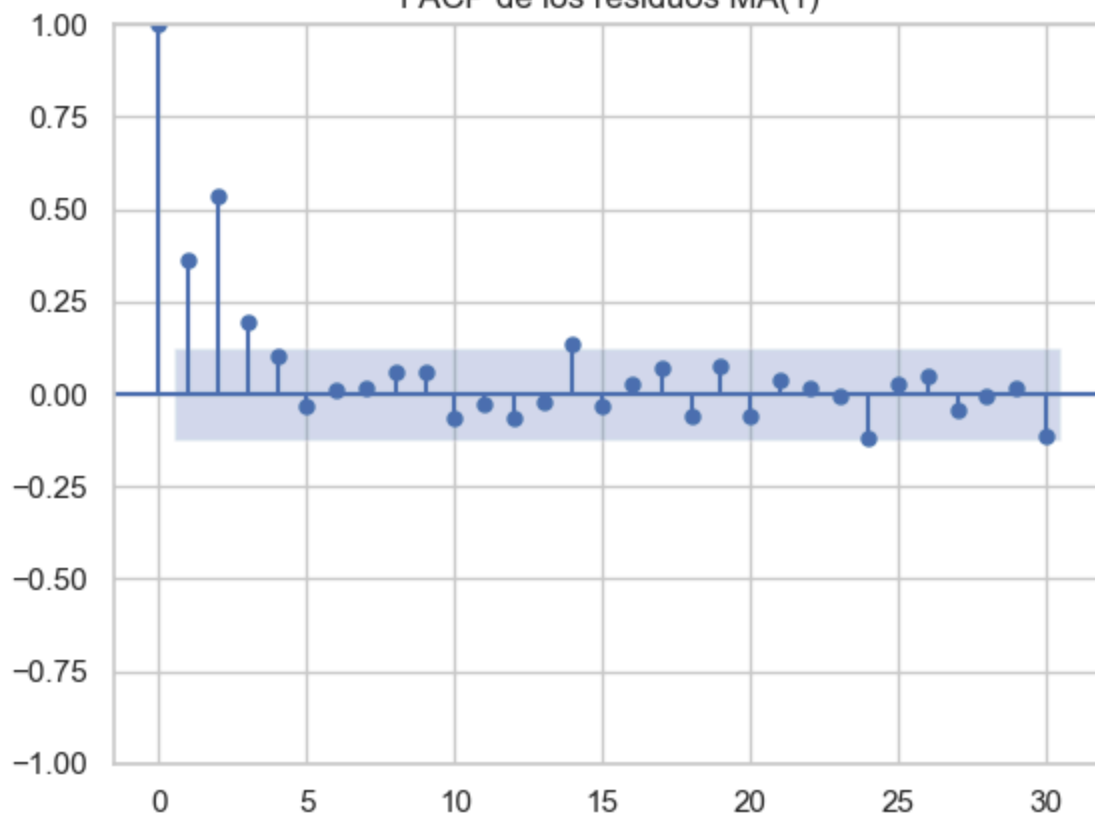


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<Figure size 800x400 with 0 Axes>

FACP de los residuos MA(1)



=== Modelo ARMA(1,1) estimado sobre la serie en niveles ===

SARIMAX Results

```
=====
Dep. Variable:    Tasa_participacion    No. Observations:    249
Model:            ARIMA(1, 0, 1)        Log Likelihood        -335.943
Date:             Sat, 15 Nov 2025      AIC                    679.885
Time:             18:50:46              BIC                    693.955
Sample:           01-01-2005            HQIC                   685.549
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
const         43.4391     0.620     70.053     0.000     42.224     44.654
ar.L1          0.8998     0.039     23.089     0.000      0.823      0.976
ma.L1         -0.1875     0.053     -3.560     0.000     -0.291     -0.084
sigma2         0.8651     0.027     31.734     0.000      0.812      0.919
=====
Ljung-Box (L1) (Q):           0.16    Jarque-Bera (JB):           41584.32
Prob(Q):                     0.69    Prob(JB):                   0.00
Heteroskedasticity (H):       4.28    Skew:                      -5.50
Prob(H) (two-sided):         0.00    Kurtosis:                   65.35
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Parámetros principales:

Constante: 43.4391

Phi_1 (AR1): 0.8998

Theta_1 (MA1): -0.1875

AIC: 679.89

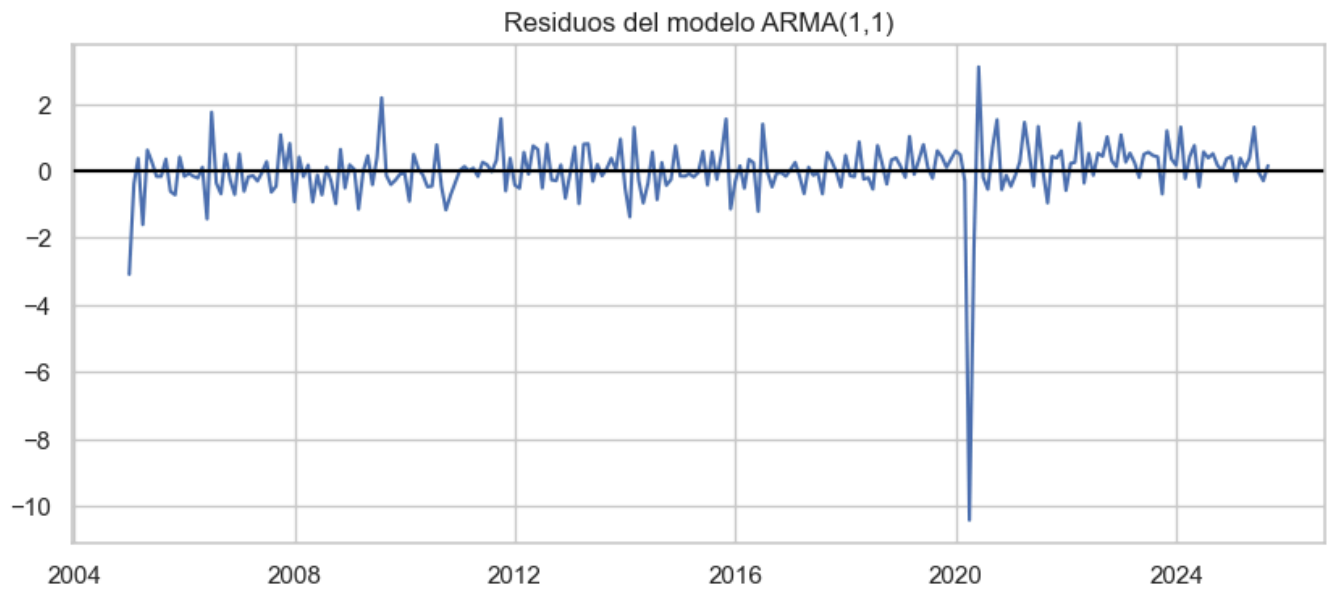
BIC: 693.96

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
```

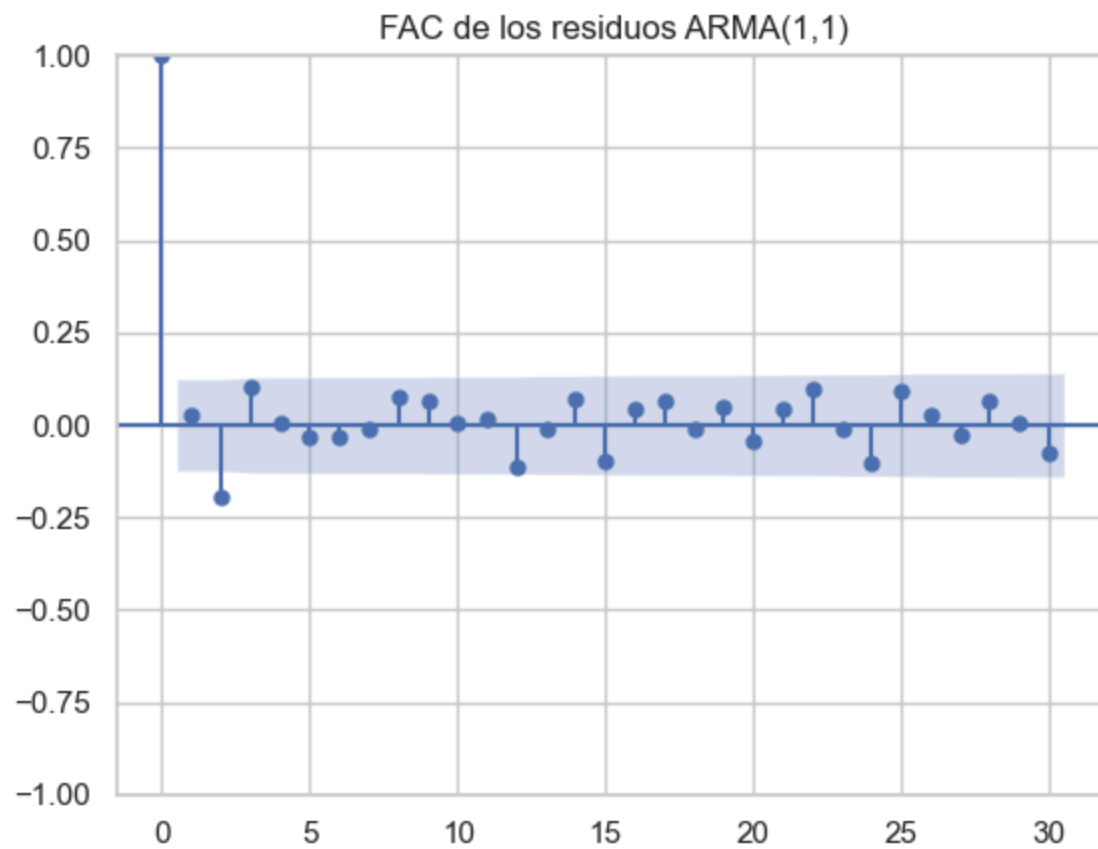
=== Prueba de Ljung-Box (residuos ARMA(1,1)) ===

```
      lb_stat  lb_pvalue
10  15.931045   0.101623
```

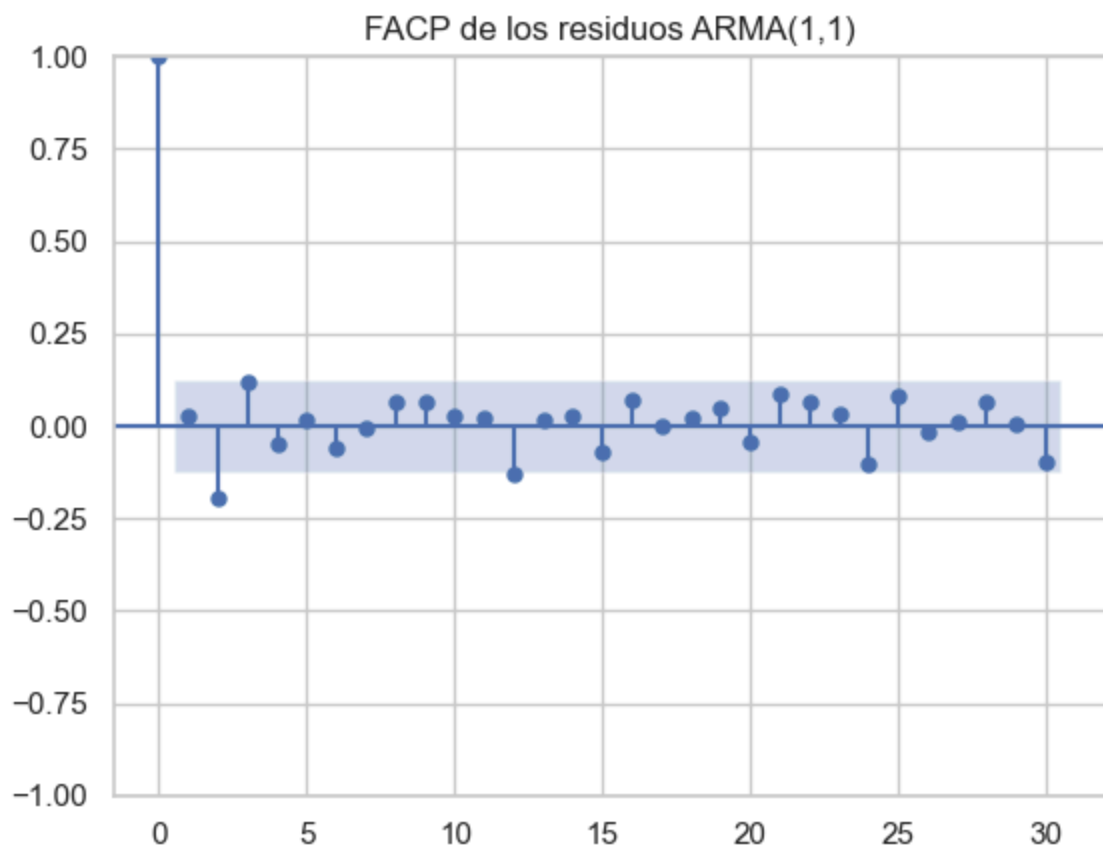
Conclusión: No se rechaza H_0 de 'no autocorrelación' en los residuos (ruido blanco).



<Figure size 800x400 with 0 Axes>



<Figure size 800x400 with 0 Axes>



	Modelo	AIC	BIC
0	AR(1)	681.295796	691.848155
1	MA(1)	805.658408	816.210766
2	ARMA(1,1)	679.885451	693.955262

	Modelo	Parametro	Estimacion	p_value	Significativo_5pct
0	AR(1)	const	43.449419	0.000000e+00	True
1	AR(1)	ar.L1	0.848004	3.495771e-164	True
2	MA(1)	const	43.472801	0.000000e+00	True
3	MA(1)	ma.L1	0.745530	2.542017e-83	True
4	ARMA(1,1)	const	43.439123	0.000000e+00	True
5	ARMA(1,1)	ar.L1	0.899776	5.984190e-118	True
6	ARMA(1,1)	ma.L1	-0.187464	3.701498e-04	True